

AHMED FOUAD LAGHA

PhD Researcher @ ELTE | Data Science & ML

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SUMMARY

Dedicated PhD Researcher specializing in Privacy-Preserving Machine Learning and Data Science. Proven track record in developing state-of-the-art Deep Learning models and implementing secure AI systems using Federated Learning and Differential Privacy. Experienced in software engineering with a strong background in architecting scalable web applications and LLM-based orchestration pipelines. Committed to advancing the frontier of secure and trustworthy artificial intelligence.

EXPERIENCE

Teaching Assistant

Feb. 2026 – Jun. 2026

Eötvös Loránd University (ELTE)

Budapest, Hungary

- Designed practical labs focusing on Adversarial Machine Learning and Privacy-Preserving AI using PyTorch.
- Guided students through the implementation of state-of-the-art attacks, including Evasion (FGSM, PGD), Data Poisoning (Backdoors), and Membership Inference.
- Mentored students on defensive strategies like Differential Privacy (DP-SGD), Federated Learning, and Adversarial Training. Course Link: [HERE](#)

PhD Researcher (Privacy-Preserving ML)

Sep. 2025 – Present

Eötvös Loránd University (ELTE)

Budapest, Hungary

- Implemented federated training flows and applied Differential Privacy techniques for synthetic data generation.
- Built evaluation pipelines to quantify privacy-utility tradeoffs and documented results demonstrating improved privacy guarantees with acceptable predictive performance.
- Awarded the Stipendium Hungaricum Scholarship based on academic excellence and research potential.

Software Engineer Intern

Jun. 2023 – Feb. 2025

ALX Africa

Remote

- Developed design systems and analyzed project limitations, performance requirements, and interfaces to streamline development.
- Modeled applications and system architectures to assist developers in writing efficient and maintainable code.
- Evaluated system capabilities and requirements through data manipulation, storage, and retrieval optimization.
- Collaborated in a remote environment, utilizing project management and creative problem-solving skills to meet technical objectives.

PROJECTS

MyVellum | LLM Orchestration, RAG, Docker, CI/CD

Dec. 2025 – Present

- Designed and implemented LLM orchestration and modular prompt templates to standardize model usage across product teams.
- Built vector-backed RAG pipelines and context-stitching services to ground model responses and reduce hallucinations in production.
- Integrated automated evaluation gates into CI/CD to track relevance and safety metrics for model updates.
- Owned API design and Docker-based deployment pipelines; mentored two engineers in prompt engineering and testing best practices.

RayaneBio E-Commerce Platform 🌐 | React, Next.js, HTML, CSS

Dec. 2024 – Mar. 2025

- Architected and deployed a responsive e-commerce platform for organic products resulting in seamless user experiences and increased client satisfaction.

- Implemented advanced product display, secure user authentication, and intuitive navigation features, enhancing customer engagement through integrated testimonials.
- Designed and showcased company values and customer testimonials, driving brand loyalty and positive user feedback.

Discretuna Text Summarizer 🐍 | *Python, Flask, NLTK, scikit-learn, HTML, CSS* May 2025

- Developed a web application for automatic text summarization using the Discrete Tuna Optimization (DTO) algorithm.
- Implemented features for text input/file upload and adjustable summarization parameters.
- Provided statistical analysis of summary quality, including ROUGE score and readability metrics.

E-Learning Platform 🐍 | *Node.js, Express.js, MongoDB, React, Redux, SCSS* Jul. 2025

- Designed and implemented RESTful APIs for user authentication, course management, and content delivery.
- Developed responsive UI components using React and Redux for an engaging learning experience.
- Integrated secure user authentication and authorization mechanisms.

WAU-Net 🐍 | *Python, TensorFlow, Keras, OpenCV* Jan. 2025 – Jun. 2025

- Engineered and trained a high-performance deep learning model for hair segmentation using Python, TensorFlow, and Keras, achieving state-of-the-art accuracy on benchmark datasets.
- Designed a comprehensive data augmentation pipeline, significantly improving model generalization and robustness across diverse image conditions.
- Evaluated model efficacy with rigorous metrics, attaining 95.37% Dice coefficient for efficient real-time deployment.

EDUCATION

Eötvös Loránd University (ELTE) <i>Ph.D. in Informatics</i>	Sep. 2025 – Expected 2029 <i>Budapest, Hungary</i>
University of Khenchela <i>M.Sc. in Artificial Intelligence, GPA: 3.7/4.0</i>	Sep. 2023 – Jun. 2025 <i>Khenchela, Algeria</i>
University of Khenchela <i>B.Sc. in Mathematics and Computer Science, GPA: 3.9/4.0</i>	Sep. 2020 – Jun. 2023 <i>Khenchela, Algeria</i>

CERTIFICATIONS

- **AI Career Essentials (AiCE) certification** – ALX South Africa (Jul. 2024)
- **Introduction to Deep Learning with PyTorch** – Datacamp (Feb. 2025)
- **ALX Software Engineering with a specialization in Back-end** – ALX Africa (Feb. 2025)
- **Advanced Learning Algorithms** – DeepLearning.AI, Stanford University (Apr. 2024)
- **Supervised Machine Learning** – DeepLearning.AI, Stanford University (Apr. 2024)

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, Bash, JavaScript

Databases: MySQL, SQLite, MongoDB

Data Science: Pandas, NumPy, Matplotlib, Seaborn

Machine Learning: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV

Web Development: HTML, CSS, Flask, Node.js

Tools: Git, Linux, CI/CD

Privacy-Preserving ML: Federated Learning, Differential Privacy

LANGUAGES

Arabic: Native | **English:** Professional Proficiency | **French:** Intermediate